

Reproducibility without a mandate: computational reproduction and multiverse analysis of exercise, rehabilitation and physiotherapy trials with publicly deposited individual participant data

Jürgen Degenfellner

Institute of Physiotherapy, School of Health Sciences,
ZHAW Zurich University of Applied Sciences, Winterthur, Switzerland
degn@zhaw.ch ORCID 0000-0001-6886-4583

Draft, 13 September 2026

Abstract

Background. Large reproduction projects have recomputed published results from the original data and code, but only in fields where journals require authors to deposit both. Journals in physiotherapy and rehabilitation ask at most for a data availability statement. None requires deposition of data and code as a condition of publication. I asked how many trials in these fields a reader can recompute from public data deposits without contacting the authors, whether the published estimates reproduce, and how far they move when the analysis is varied over choices classified in advance as preserving or as changing the estimand.

Methods. From 423 open-access records in Europe PMC I identified 152 studies with a repository link in their availability statement, retrieved the deposits, and assessed the 48 randomised trials with a readable data file under fixed rules for identifying the primary outcome, control arm and published model, with trial registrations retrieved. I fitted each published primary analysis as described and ran a multiverse over a grid of decision nodes written before computation: an estimand-preserving grid (baseline handling, covariates, model family, inference) and a defensible grid adding missing-data strategy and transformation. After inspecting the first results I added a strict grid with baseline handling fixed to analysis of covariance and moved outlier rules, which no included paper uses, to a sensitivity arm. Effects are in pooled within-group standard deviations.

Results. Sixteen trials were included, 11% of those with a link. Of 136 losses, 104 occurred before the data were opened. Of the 32 after, 11 were deposits that could not support the published analysis. Fitting the published analysis recovered 11 of 12 point estimates and all 12 to 0.014 standard deviations. Over 196 estimand-preserving specifications the published result moved by at most 0.15 standard deviations in the median trial and by 0.41 at most. No specification reversed its sign. In 11 of 16 trials only baseline handling could vary. The displacement scaled with baseline imbalance (Spearman 0.82). With baseline handling fixed, the range was zero in 12 trials and at most 0.04 in the rest. The defensible grid widened the range by a median factor of 1.35, mostly through logarithmic transformation. It reversed the published sign in single specifications of two trials with effects near zero. Three of 12 primary reports do not identify a primary outcome through registration or paper. Four papers report no point estimate. My first mapping of analyses onto data reversed the effect

direction in 7 of 16 trials before rules were fixed.

Conclusions. In a field whose journals do not require data deposition, 16 of the 152 trials that pointed to a repository, about one in nine, could be recomputed by a reader without contacting the authors. Where the published analysis could be fitted as described, it reproduced. Over choices that keep the estimand, a multiverse on simple two-arm trials has little to vary beyond the handling of the baseline value, whose effect is set by baseline imbalance. Most of what a wider multiverse adds is a change of question.

1 Introduction

Computational reproduction of published research at scale has been demonstrated where journals require authors to deposit data and code. Brodeur and colleagues reproduced 110 articles from economics and political science journals with mandatory data and code policies and found more than 85% of them computationally reproducible. Of the estimates the original authors called “statistically significant”, 72% remained “significant” with the same sign under robustness checks chosen by the reproducers [1]. A policy that is not verified at submission does less: under the data and code policy of Science, artefacts could be obtained for 44% of a sample of articles and results reproduced for 26% [2]. Biomedical audits have stalled one step earlier. Naudet and colleagues identified 37 randomised trials published under the data-sharing policies of the BMJ and PLOS Medicine, obtained data for 17, reproduced the primary result fully in 14 and reached the same conclusion in 16 [3]. For trials assessed by the European Medicines Agency, Siebert and colleagues could reanalyse 10 of 62 (16%) [4]. In psychology, about a third of the datasets shared under the mandatory policy of Cognition were not reusable [5]. Among articles with reusable data and an open data badge at Psychological Science, the obstacle to reproduction was unclear reporting of the analysis [6].

Physiotherapy and rehabilitation have no data-sharing mandate. Elghzali and colleagues counted data availability statements in 1 278 articles from the five rehabilitation journals with the highest impact factors: 25.5% of articles carried one. The figure depends on journal policy, because one journal that requires a statement had one in 99% of its articles and the other four together in 5%. A statement is not a deposit. None of these journals requires deposition of data or code [7]. In the journals of four national physiotherapy associations, 12% of 465 articles made data directly available. The authors recommend a default open data policy that the field does not yet have [8]. When authors of rehabilitation studies whose availability statements promised data on request were asked, 22.7% complied [7]. In sports and exercise science, 14% of authors provided raw data on request. Of attempted replications, 28% succeeded [9]. Dhanani and colleagues recomputed the statistics of 572 trials from the numbers printed in the articles, without individual data [10]. To the best of my knowledge, no published or registered study in physiotherapy, rehabilitation or exercise science has retrieved individual participant data from public repositories and recomputed the published primary analysis. I searched Europe PMC, OpenAlex and the OSF registry index on 15 September 2026 for combinations of reproducibility, reanalysis, multiverse, specification curve and many-analysts terms with terms for these fields and for individual, raw or shared data; the queries and their results are documented in the Supplement. The nearest neighbour is a registered report from the German Sport University Cologne that plans to recompute the primary analyses of 50 articles in the Journal of Sports Sciences from raw data shared privately by the authors [11], alongside a preprint by the same group on data and code availability in sports science [12]. Reproducibility in this literature usually means whether a clinician can reproduce an exercise programme from its description. I mean computational reproducibility, the recovery of the published number from the published data and methods.

A recomputed estimate answers one question: was the reported number obtained by the reported method. A second question is how far the number depends on choices that the method leaves open. Multiverse analysis and specification curve analysis address this question by computing every defensible combination of analytical choices [13, 14]. The method is popular and loosely applied. Of 613 multiverse-

style studies, the median study computed 144 specifications. Among the 152 coded in depth, 8.6% were preregistered, 3.9% discussed whether their specifications were equivalent or defensible, and 82.2% reported the results descriptively without any inferential summary [15]. Del Giudice and Gangestad showed that a multiverse padded with options that are not equivalent can obscure a real effect. They proposed classifying each decision into three types. A Type E decision has options of comparable validity that examine the same effect and estimate it with comparable precision. A Type N decision has one option with justified precedence, or options that answer different questions. A Type U decision is one where equivalence is uncertain [16]. Short and colleagues separate the defensibility of a single pipeline from the equivalence of pipelines with one another, and warn that a multiverse whose pipelines produce nearly identical datasets creates an illusion of consistency [17]. In a simulated behavioural trial, Veltri found that preprocessing choices explained 76.9% of the variance of the effect estimate and model choices 7.5%, but the simulation did not separate choices that change the estimand from choices that do not [18]. Crenshaw and colleagues applied the multiverse approach to one clinical trial [19]. The epidemiological version of the same question, vibration of effects, has been studied for observational associations [20, 21]. Studies in which many analysts receive the same data have shown how far conclusions vary between competent analysts [22–25]. None of these applied one prespecified, equivalence-classified grid to a corpus of trials.

I asked four questions. First, of trials in exercise, rehabilitation and physiotherapy with publicly deposited data, in how many can the published primary analysis be identified in the data and recomputed by a reader who contacts no one? Where is the loss? Second, does the published primary estimate reproduce? Third, over a prespecified grid of choices classified as equivalent, how far does the standardised effect move, and which choice moves it? Fourth, how much range does a wider grid of defensible but non-equivalent choices add? Through which choices? I also record which reporting features of the papers obstructed reproduction and report how often my own first mapping of published analyses onto deposited data was wrong before I fixed rules for it.

2 Methods

2.1 Design and transparency

I conducted a meta-research audit of randomised trials in exercise, rehabilitation and physiotherapy that had deposited individual participant data in a public repository. The audit has three stages. Stage 1 asks in how many trials the published primary analysis can be identified in the deposited data and recomputed without contacting anyone. Stage 2 asks whether the published primary estimate reproduces. Stage 3 applies the same prespecified grid of analytical choices to every trial and measures how far the estimate moves.

The audit was not preregistered. The grid of decision nodes was written on 28 August 2026, before any specification was computed. The rules that map a published analysis onto the deposited data (section 2.3) were fixed after I had extracted the published analyses from all candidate papers and before the final computation. Three changes to the grid were made after the first complete run had been inspected. The strict grid of section 2.6 was added. Outlier rules, which had been a core node, were moved to a sensitivity arm after a winsorising rule on a bounded outcome had produced a range of 1.08 standard deviations in one trial. The core grid was renamed from principled to estimand-preserving once the precision criterion

of Del Giudice and Gangestad had been applied to baseline handling. All three are stated where they matter. All results were inspected before this manuscript was written. Brodeur et al. state that their own reanalyses “should thus all be considered as not pre-registered” because the implementable checks cannot be known before a replication package is opened [1]. The same applies here. Section 3.5 reports what the rule-based mapping changed relative to my first, ad hoc mapping. The transparency items proposed by van den Akker et al. for secondary data analysis are addressed in Supplement Text S1 [26]. Code, the trial registry, the grid definition and all outputs will be deposited in a public repository on acceptance.

2.2 Corpus

I searched Europe PMC on 27 August 2026 for open-access articles with physiotherapy, rehabilitation or exercise-therapy terms in the title or abstract, a clinical design (publication type randomised controlled trial, clinical trial or observational study, or corresponding title terms), and a repository term anywhere in the full text. The search returned 423 records. For each record I retrieved the JATS full text and extracted the data- and code-availability statement. A record was retained when a repository URL appeared inside that statement. Counting URLs anywhere in the full text had inflated an earlier pass with citations of software repositories. Restricting the count to the availability statement removed most apparent matches, which had been citations of software repositories rather than deposits. The restricted count gave 152 studies, 91 of them randomised trials. Data supplied only as a journal supplementary file, without a repository link, were not captured by this route.

For every retained study I queried the application programming interfaces of OSF, Zenodo, Dryad and figshare, listed the deposit and classified files by extension. A deposit counted as reachable when a file list was returned and as containing data when at least one tabular file (csv, xlsx, sav, rds or similar) was present. The deposit probe left 84 reachable deposits, 74 with a tabular file, 48 of them randomised trials. These 48 were downloaded and assessed by hand.

A second search route, querying Zenodo and Dryad directly for clinically relevant deposits, returned 127 study-like deposits. These were not matched to articles for this report and are discussed as an extension in section 4.7.

2.3 Eligibility and mapping rules

A trial entered stages 2 and 3 when four conditions held. It was a randomised trial of an exercise, rehabilitation or physiotherapy intervention. Individual participant data were downloadable without application or data-use agreement. The primary outcome and the allocation were identifiable in the data. The published methods described the primary analysis in enough detail to reconstruct it. Group-level summary tables did not count as individual data. I did not contact authors.

Design exclusions were applied to all 48 trials in the same way. Crossover and within-person designs were excluded because the grid assumes parallel groups. A binary primary outcome was excluded because the grid is built for continuous outcomes. The other exclusions were feasibility studies whose primary aim was recruitment or retention, studies without a clinical or training intervention, non-randomised studies and secondary analyses of an included trial that report no arm contrast. Every exclusion is recorded with its category and reason in Supplement Table S1.

Mapping a published analysis onto deposited data requires decisions about which variable is the primary outcome, which timepoint is primary, which value of the group variable is the control arm, and which covariates entered the published model. My first mapping made these decisions study by study. After the published analyses had been extracted from all 21 candidate papers with verbatim quotations, I fixed eight rules and rebuilt the mapping under them. The primary outcome is the one named in the trial registration, retrieved from the registry for every trial with a registration identifier. Failing that, it is the one the paper calls primary, then the outcome of the sample-size calculation, then the first outcome reported in the abstract. The source is recorded for each trial. A paper whose headline outcome is not the registered primary outcome of its trial is a secondary report. The primary timepoint is the one the paper designates, otherwise the first follow-up after the end of the intervention. When the primary outcome or its timepoint is absent from the deposit, the trial is excluded. No substitute outcome is used. The control arm is the arm the paper describes as control, placebo, sham, wait-list or usual care. Its code in the data is verified through value labels or, when labels are absent, by matching group sizes and baseline means against the paper's first table. Covariates are those that appear as adjustment terms in the published primary model. Variables tested only for baseline balance do not count. Each paper is classified as a primary report, which reports the trial's prespecified primary analysis, or a secondary report, which presents an exploratory or secondary analysis with its own headline estimate. Field and population are coded for each trial. The mapping was done by one coder.

2.4 Stage 2: computational reproduction

From each included paper I extracted the published primary estimate, its 95% confidence interval (a credible interval for the one Bayesian analysis) and its p-value, and the specification of the published analysis (baseline handling, covariate set, analysis population, missing-data strategy, model family, standard-error method, transformation), each with a verbatim quotation. Where a paper reported only a test statistic or a p-value, the trial was classified as having no point estimate. Two documented adjustments were made to the extracted values. For the time-restricted eating trial [27] the published primary estimate is the time-by-group interaction of a mixed model, reported as control minus intervention (2.13 kg). All arm contrasts in this paper are intervention minus control, so the value enters as -2.13 kg with its interval reflected accordingly. For the phosphatidic acid trial [28] the value first extracted was an F statistic. The trial was reclassified as having no point estimate.

For each of the 12 trials with a point estimate I then fitted the published analysis as the paper describes it, independently of the multiverse grid. Six trials required more than a single linear model or an assumption the paper leaves open. For the cluster-randomised FREE trial [29] the published analysis was a linear mixed model with random intercept and slope for practitioner and the prespecified covariates present in the deposit. For the balance feedback trial [30] it was a mixed model over all four measurement occasions on the logarithmic scale, with the authors' deposited code. TEREKO [31] was analysed with a constrained longitudinal analysis. The deload trial [32] was fitted with a Bayesian multilevel model with default priors, because the paper states none. The water exercise trial [33] was analysed with multiple imputation with 20 imputations, because the paper does not state the number. For the frozen shoulder trial [34] it was a repeated-measures analysis of covariance over the two short-term follow-ups, approximated as a mixed model. Where a detail was not stated, the script records the assumption. Agreement was classified by the

rule fixed before any computation: full when the recomputed estimate had the same sign and differed by at most 2% from the published one, or by less than half a unit of the last printed digit, a minor discrepancy at most 10%, a major discrepancy otherwise [3, 6]. The absolute difference in pooled standard deviations of the raw outcome, whether the recomputed 95% interval contains the published estimate, and whether the p-value falls on the same side of 0.05 are reported for every trial without thresholds. I did not judge whether the published analysis was appropriate.

Separately, to locate the published estimate in the multiverse of section 2.6, each published specification was matched to the grid specification agreeing with it on the largest number of nodes. Published model families absent from the grid counted as non-matches. Ties were broken by the estimate closest to the published one and are reported. The percentile of the published estimate in the distribution of the estimand-preserving grid is given where both are on the same scale.

2.5 Data integrity screen

Before any multiverse was computed, each dataset was screened for patterns that would make a fragility analysis meaningless. For every continuous baseline variable named in the paper I computed a one-way analysis of variance p-value for the difference between arms. Under correct randomisation these p-values are uniform [35]. They were combined by Stouffer's method, which suits a two-sided question better than Fisher's combination because arms that are too similar and arms that are too different are both of interest. A combined value below 0.005 or above 0.995 was treated as a triage flag, not as a verdict. The test was applied to continuous variables only. Bolland et al. showed that rounding distorts the distribution of these p-values [36]. The deposited baseline variables are rounded to varying degrees, which I did not correct. Duplicate rows over all numeric columns were counted. A terminal-digit test on the outcome was computed but not used for flagging, because several primary outcomes are derived means whose terminal digits are structurally non-uniform. Flagged trials would have been excluded from stage 3 and reported separately. The named-variable component requires at least two continuous baseline variables named in the paper and present in the deposit. A broader search over all deposited variables whose names indicate a baseline measurement was computed for every trial with such variables and is reported in the Supplement, but was not used for flagging because it includes variables the paper did not treat as baseline characteristics.

2.6 Stage 3: the grids

The multiverse varies six decision nodes (Table 5). Following Del Giudice and Gangestad, each node was classified before computation as Type E, Type N or Type U. A Type E node has options of comparable validity that examine the same effect and estimate it with comparable precision. A Type N node has one option with justified precedence, or options that examine different effects. A Type U node is one where equivalence is uncertain [16]. I depart from this scheme in one respect and say so. The three ways of handling the baseline value, follow-up value, change score and analysis of covariance, estimate the same mean difference under randomisation but not with the same precision: analysis of covariance is more efficient than the change score, which is more efficient than the follow-up value when baseline and outcome are correlated [37–39]. Under a strict reading, precedence goes to analysis of covariance and the

node is Type N. I kept the node in the core grid because none of the three changes the quantity estimated, which is the criterion the grid is named for. I call that grid estimand-preserving. The strict grid without it shows what the precision criterion removes.

Three grids are reported. The estimand-preserving grid varies baseline handling, the covariate set (none, the stratification factors, or the published set) [40, 41], the model family (linear model, or a random intercept for a cluster variable where one is estimable) and the inference method (model-based standard error, heteroscedasticity-consistent HC3 standard error, or a permutation test with 4 999 permutations). The strict grid fixes baseline handling to analysis of covariance where a baseline exists and varies the remaining three nodes. The defensible grid, in the sense of Short et al. [17], adds two Type N nodes. The first is the missing-data strategy. Its options are complete cases, last observation carried forward where a baseline exists, or multiple imputation by predictive mean matching with 20 imputations, Rubin's rules and Barnard and Rubin degrees of freedom. This node is Type N on both criteria, because complete cases and imputed data target different populations once dropout is informative and because imputation has justified precedence under a missing-at-random assumption [42]. The second is logarithmic transformation of outcome and baseline, which estimates a ratio of geometric means. The difference between the defensible and the estimand-preserving grid is a result of the audit, not a robustness check.

Outlier rules were part of the original grid as a Type E node with pooled winsorising at 3 standard deviations, 1.5 interquartile ranges or 3 median absolute deviations. No paper in the corpus reports an outlier rule, so the node fails the criterion of use in the literature. Winsorising also changes the quantity estimated. The node was reclassified as Type U, removed from the three grids, and computed as a separate sensitivity arm, which is reported because it illustrates what an unclassified option can do to a bounded outcome.

An option was dropped for a trial when it could not change the data: missing-data options without missing outcomes, cluster models without an estimable cluster variable, logarithms of non-positive values, last observation carried forward without a baseline. For trials deposited in long format the baseline was taken from the row at the baseline timepoint. HC3 and permutation inference are defined only for the linear model, and permutation is not combined with multiple imputation. Multiple imputation used one set of imputed datasets per trial and transformation, shared by all specifications, so that Monte Carlo variation between imputation runs does not count as specification variation. Specifications that returned identical estimate, standard error and p-value were collapsed to one. The share of collapsed duplicates is reported. Non-applicable nodes and the number of nodes that vary are counted per trial, because a share of 100% for one node in a trial where no other node can vary is not a finding.

2.7 Summary measures

Estimates from different specifications are comparable only on a common scale. Every estimate was divided by the pooled within-group standard deviation of the raw outcome at the primary timepoint, computed once per trial before any preprocessing, which gives a standardised difference d in the convention of Cohen's d and Hedges' g . The residual standard deviation of the fitted model was not used, because it changes with the specification and would manufacture range. Specifications on the logarithmic scale were standardised with the pooled standard deviation of the logarithmic outcome and were never pooled with raw-scale specifications.

The primary summary is the displacement of the published result. For each trial with a point estimate, the estimate recomputed in stage 2 is standardised on the same scale as the grid and taken as the anchor. Every specification is expressed as the difference between its d and the anchor. Per trial and grid I report the median, the range and the largest absolute displacement, the share of specifications within 0.1 standard deviations of the anchor, and whether any specification reverses the sign of the published estimate. Trials whose paper reports no point estimate have no anchor and are described by the distribution of d alone. As secondary description, for each trial and grid I report the median, interquartile range and range of d over unique specifications, together with the share of specifications whose sign agrees with the median. I also report the share with $|d|$ above 0.2, Cohen’s convention for a small effect, since instrument-specific minimal important differences were not available for every outcome. The s -value, the negative base-2 logarithm of the p -value, is given as a continuous measure of evidence against the test hypothesis [43], as is the percentile of the published estimate in the distribution of d . The permutation p -value cannot fall below 1 in 5 000, which caps the s -value of permutation specifications at 12.3 bits. I do not report the share of “statistically significant” specifications as a primary measure [44].

To identify which node moves the estimate, I fitted for each trial and grid an additive linear model of d on all varying nodes and partitioned the explained variance by Type II sums of squares, which do not depend on the order of terms in an unbalanced grid. The same partition was computed for the standardised width of the 95% interval, because inference options change the interval but not the estimate. Removing incoherent combinations confounds some node pairs, in particular model family with inference method. As a sensitivity analysis I refitted the model with all two-way interactions in the ten trial-by-grid combinations with at least 30 specifications and checked whether the node with the largest share changed. The number of trials in which only one node can move the estimate is reported, because a share of 100% in such a trial is not a finding.

Empirical similarity of the preprocessed datasets was measured by the Pearson correlation between the outcome vectors of every pair of preprocessing pipelines of a trial. Where all pairs exceed 0.99, the pipelines barely change the data. Stability of the sign is then not evidence of robustness [17]. Descriptive correlates of the principled range (sample size, standardised baseline imbalance, baseline-outcome correlation, share of missing outcomes) are given as Spearman coefficients over 16 trials without tests.

Proportions are reported with Wilson 95% intervals. Medians over trials are reported with 95% intervals from 5 000 bootstrap resamples of trials, not of specifications. Confidence intervals of the published estimate and of any multiverse summary were not compared for overlap. Pooled estimation over specifications [45] and multiplicity control over specifications [46] were not performed.

Analyses used R 4.6.0 with the packages `dplyr`, `readxl`, `haven`, `lme4`, `lmerTest`, `sandwich`, `lmtest`, `mice` and `car`.

3 Results

3.1 Feasibility

Figure 1 shows the corpus.

Of 423 records, 152 had a repository link inside the availability statement (91 randomised trials). Two

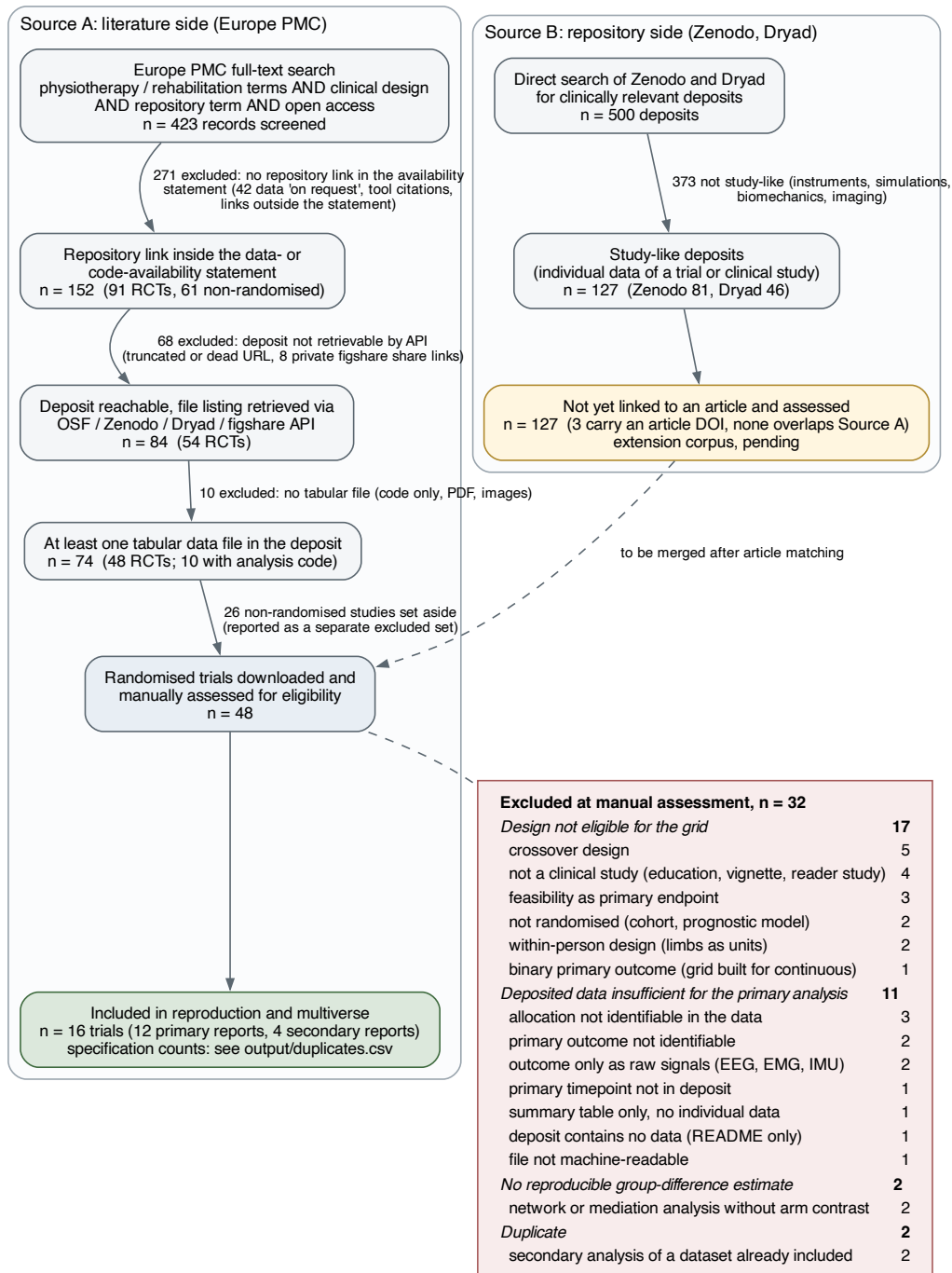


Figure 1: Corpus flowchart. Source A is the literature route through Europe PMC; source B the repository route through Zenodo and Dryad, not merged for this report. The first loss (152 to 48) occurs before the data are opened; the second (48 to 16) after. Exclusion categories from `config/exclusions.csv`.

losses followed. The first is the availability loss. In this step 68 deposits could not be retrieved through the repository interfaces, mostly because of truncated or dead links. A further 10 reachable deposits contained no tabular file. Of the remaining 74 studies, 26 were not randomised. This left 48 randomised trials with a readable data file, 32% of those with a link. The second loss occurred after the data had been opened. Of the 48 trials, 17 were excluded for design (crossover 5, not a clinical or training intervention 4, feasibility as the primary aim 3, not randomised on inspection 2, within-person allocation of limbs 2, binary primary outcome 1), 11 because the deposited data could not support the published primary analysis (allocation not identifiable 3, primary outcome not identifiable 2, outcome present only as raw electrophysiological or inertial signals 2, primary timepoint absent 1, summary table instead of individual data 1, README without data 1, file not machine-readable 1), 2 because the paper reported no arm contrast (a network analysis and a mediation analysis), and 2 as secondary analyses of a dataset already included. In 11 trials the deposit could not support the published analysis. In eight of them the data were individual and readable.

Sixteen trials entered stages 2 and 3, 33% of the 48 assessed (Wilson 95% interval 22 to 47%) and 11% of the 152 with a link (7 to 16%). Twelve are primary reports. Four analyse an outcome that the trial registration lists as secondary (an exploratory balance analysis [30], a psychological outcome of a pregnancy exercise trial [47], postpartum depression in a water exercise trial [33], and voluntary muscle activation in a neuromodulation trial [48] whose registered primary outcome, muscle strength, is not in the deposit). Deposits were hosted on OSF (7), Zenodo (6) and Dryad (3). Two contained analysis code alongside the data. Six trials are physiotherapy or rehabilitation trials in patients or survivors [29, 31, 49–52], one a neuromodulation trial in patients [48], one a corticosteroid injection trial for frozen shoulder [34], six training or nutrition trials in healthy adults [27, 28, 30, 32, 53, 54], and two exercise trials in pregnancy [33, 47]. Eight interventions were delivered by physiotherapists. The trials randomised 18 to 404 participants. The number analysable for the primary contrast in the estimand-preserving multiverse ranged from 16 to 276. The fitted published models used 20 to 373 observations.

3.2 Reproduction

Twelve of the 16 papers report a point estimate for the primary arm contrast. Fitting the published analysis as described recovered it fully in 11 (92%, Wilson interval 65 to 99%) and with a major discrepancy in 1 (Table 5). The absolute difference between published and recomputed estimate had a median of 0.001 standard deviations and a maximum of 0.014. Ten estimates agree with the published value to the precision at which it was printed. The one major discrepancy is the deload trial [32] (18%, 0.014 standard deviations), whose published Bayesian multilevel model states neither priors nor seed. My fit with default priors gives -0.51 against -0.63 , with credible intervals that overlap almost entirely. In all 12 trials the recomputed 95% interval contains the published estimate. In the 8 trials that report a p-value and for which a frequentist p-value exists, the recomputed value falls on the same side of 0.05. One paper did not report an interval for the primary estimate and three did not report a p-value. The test statistics of the four papers without a point estimate are treated below. Four of the 12 reconstructions rest on an assumption the paper leaves open: the number of imputations in the water exercise trial [33], the priors of the deload model, the estimation method in TEREKO [31], where maximum likelihood, the software default, reproduces the published interval to the decimal, and the implementation of the repeated-measures

analysis of covariance in the frozen shoulder trial [34], where the between-subject effect on the mean of the two follow-ups reproduces the published estimate to 0.1%. The FREE trial [29] reproduces to 1.8% once time is entered as a factor, as the paper describes.

For the four papers that report no point estimate, the reported test statistic was recomputed as described. The three-arm repeated-measures analysis of variance of the Mulligan trial [49] gives $F(2,40) = 0.983$, $p = 0.383$, as published. The mixed model of the cluster-set trial [53] gives the published interaction p of 0.352 and the published main- effect F statistics of 80.67 and 95.13. The exact Mann-Whitney test of the facial palsy trial [52] gives $p = 0.0020$ and the published medians and interquartile ranges of the change scores, once quartiles are computed as SPSS does. In the phosphatidic acid trial [28] the published interaction p of 0.041 reproduces, but the F statistic printed with it, $F(1,16) = 33.30$, does not: the interaction F is 4.95. No term in the analysis gives 33.30. The abstract reports the same statistic as a main effect with $p < 0.001$. The published p -values therefore reproduce in all four. One F statistic is misreported.

Counting every study with a repository link as the denominator, as Siebert et al. did for all sampled trials [4], the published primary estimate was recovered in 11 of 152 (7%, Wilson interval 4 to 12%). Counting the 48 assessed trials, it was recovered in 11 of 48 (23%, 13 to 37%).

Located in the estimand-preserving multiverse, the published estimate lies inside the range in 9 of the 11 trials on a common scale (Wilson interval 52 to 95%). The absolute difference from the multiverse median has a median of 0.016 standard deviations (bootstrap interval 0.003 to 0.056). The published specification could be represented completely in the grid in 7 of 12 trials. In 3 trials several grid specifications matched it equally well, with a range of 0.23 standard deviations between the candidates in the Lytras trial [50], whose paper does not state how the baseline entered the analysis.

3.3 Displacement of the published result over estimand-preserving choices

The estimand-preserving grid yielded 196 unique specifications over 16 trials, between 3 and 35 per trial, after collapsing 1% exact duplicates. In 11 of the 16 trials only baseline handling and the inference method could vary, because no covariate from the published model or stratification factor was in the deposit and no cluster variable was estimable. The covariate set varied in 4 trials and the model family in 2. In one trial without a baseline measurement only the inference method varied.

Figure 2 shows, for the 11 trials with a published estimate on the scale of the grid, how far each specification moves the standardised effect away from the reproduced published estimate.

The largest displacement per trial has a median of 0.15 standard deviations (interquartile range 0.08 to 0.28) and a maximum of 0.41. In 5 of the 11 trials every specification stays within 0.1 standard deviations of the published result. In none does any specification reverse its sign. The three largest displacements, 0.41 in the deload trial [32], 0.39 in the spinal cord stimulation trial [51] and 0.34 in the time-restricted eating trial [27], are each the distance between the published baseline handling and one alternative, in the three trials with the largest baseline imbalance between arms, 0.40 to 0.41 standard deviations. In the Lytras trial [50] the published follow-up analysis sits at the edge of its multiverse: the change-score and covariance-adjusted alternatives move the effect by 0.10 and 0.23 standard deviations in the same direction. Elsewhere the published estimate is central, with a median displacement of at most 0.06 in absolute value.

The distribution of d gives the same picture. Its range per trial has a median of 0.12 standard deviations

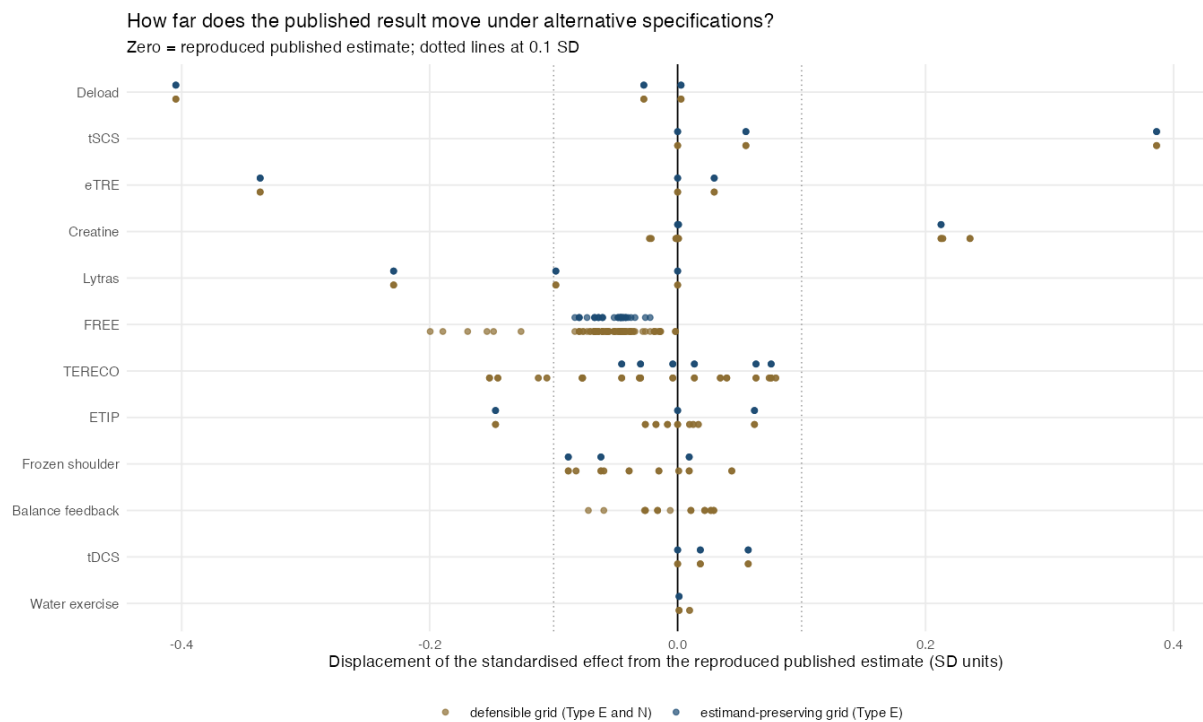


Figure 2: Displacement of the published result under alternative specifications. For each trial with a published point estimate, zero is the estimate recomputed from the published analysis, standardised by the pooled within-group standard deviation of the raw outcome; each point is one unique specification of the estimand-preserving grid (Type E nodes) or the defensible grid (Type E and N nodes). Dotted lines mark 0.1 standard deviations. The balance feedback trial appears only in the defensible grid because its published estimate is on the logarithmic scale.

(bootstrap 95% interval 0.09 to 0.21, interquartile range 0.08 to 0.22) and a maximum of 0.41. The sign of d is the same in every specification in all 16 trials (Wilson interval 81 to 100%). Supplement Figure S1 shows the median and range for every trial and Figure 3 the specification curves of two trials.

Baseline handling had the largest share of the estimate variance in 14 of the 15 trials in which any node varied (Figure 4), which in 11 of them means only that nothing else could vary.

The range is a function of baseline imbalance: the Spearman correlation between the range and the standardised difference in baseline means between arms is 0.82 over the 15 trials with a baseline and with the baseline-outcome correlation 0.39. With sample size, over all 16 trials, it is -0.29 .

The strict grid, with baseline handling fixed to analysis of covariance, leaves 68 specifications. Its range is zero in 12 of 16 trials and at most 0.04 in the other four (balance feedback [30] 0.04, FREE [29] 0.03, Tereco [31] 0.02, cluster sets [53] 0.01), the trials in which a covariate set or a cluster model could vary. Over choices that satisfy the strict equivalence criterion, there was almost nothing to vary in this corpus.

For the width of the 95% interval, baseline handling had the largest share in 11 trials, the model family in 2 and the inference method in 2. The inference method accounted for a median of 8% of the interval-width variance. Refitting with all two-way interactions in the eight trial-by-grid combinations with at least 30 specifications did not change the node with the largest share in any of them. Interaction terms accounted for a median of 13% of the explained variance (Supplement Table S4). In the eight trials in a clinical rehabilitation context the range had a median of 0.11 standard deviations and a maximum of 0.39. In the eight trials of exercise physiology or sports nutrition the median was 0.16 and the maximum 0.41. Per-trial median s -values range from 0.8 to 18 bits.

In 11 of the 16 trials every pair of preprocessing pipelines produced outcome vectors correlated above 0.99. In one trial only one pipeline existed. Sign stability in these trials says little about robustness, because the options barely change the data. The named-variable component of the integrity screen could be computed in one trial, which was not flagged. The broad search over 2 to 43 deposited baseline variables gave combined p -values between 0.15 and 0.95 (Supplement Table S6).

3.4 Estimand versus analysis

The defensible grid yielded 550 unique specifications, between 9 and 96 per trial, after collapsing 1% duplicates. Measured from the published result, the largest displacement per trial has a median of 0.18 standard deviations (interquartile range 0.08 to 0.26) and the same maximum of 0.41. Every specification stays within 0.1 standard deviations in 4 of 12 trials. In two trials single specifications reverse the sign of the published estimate, FREE and the balance feedback trial, whose published effects are 0.14 and 0.05 standard deviations. Of their specifications, 95% and 91% keep the published sign. The range of d has a median of 0.24 standard deviations (bootstrap interval 0.13 to 0.32, interquartile range 0.13 to 0.32) and a maximum of 0.52, in the spinal cord stimulation trial [51]. The sign was stable in 13 of 16 trials (57 to 93%). The range includes zero in three, the ETIP pregnancy trial [47], FREE and the balance feedback trial. Over the 15 trials with a non-zero estimand-preserving range, the defensible range exceeded it by a median factor of 1.35 (bootstrap interval 1.12 to 1.52). Baseline handling remained the node with the largest share of the estimate variance in 11 of 16 trials and the logarithmic transformation in 5. The

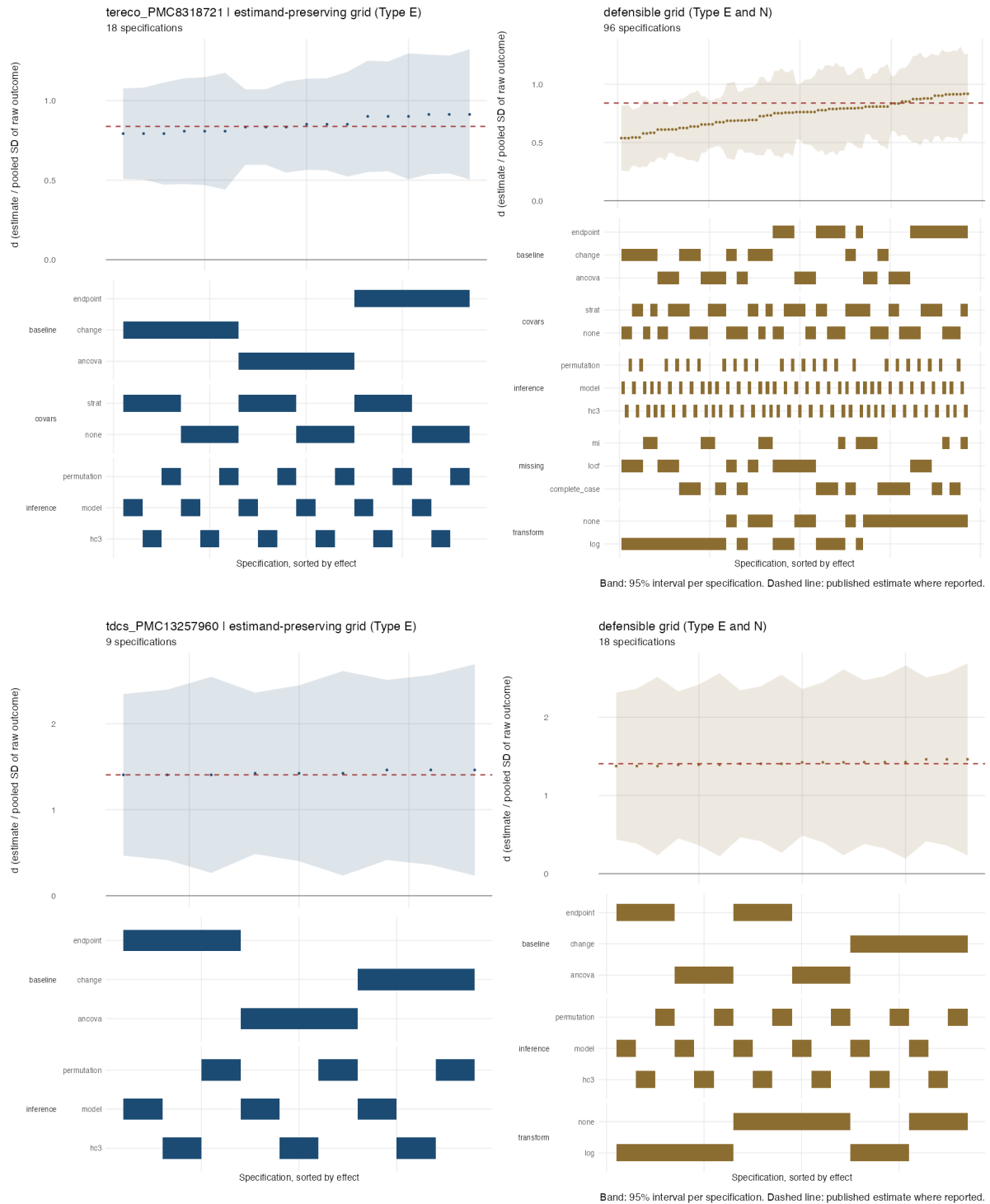


Figure 3: Specification curves for two trials: TEREKO (telerehabilitation after COVID-19, $n = 119$, top) and transcranial direct current stimulation after anterior cruciate ligament reconstruction ($n = 20$, bottom). Left panels: estimand-preserving grid; right panels: defensible grid. Points are specifications sorted by d , bands are 95% intervals, and the lower panels show the option taken at each node.

missing-data strategy, which could vary in 7 trials, had the largest share in none.

The outlier arm, which adds the three pooled winsorising rules to the estimand-preserving grid, has a median range of 0.18 and a maximum of 1.08 standard deviations. The maximum is the transcranial stimulation trial [48], whose outcome is a percentage with a ceiling at 100 and half of whose participants score between 99 and 100: the rule of three median absolute deviations winsorises the whole control arm towards the median and reduces d from 1.4 to 0.4. No trial in the corpus applied such a rule. The arm is reported as an illustration of what an option outside the literature does to a bounded outcome, not as fragility.

3.5 Reporting and the effect of my own mapping

Of the 12 primary reports, 8 analyse the outcome that the trial registration names as primary, or as the first of several co-primary outcomes (Table 5). Three trials had no registration. Of these, one declares its primary outcome in the paper, one identifies it only through the sample-size calculation, and one names none. One registered trial names no primary outcome in registration or paper. Three registrations were made after recruitment had begun. The four secondary reports state that their outcome is not the trial's primary outcome, in three cases in the paper and in one case only in the registration. Twelve of the 16 papers report a point estimate for the arm contrast. Four report only a p -value or an F statistic. One paper states an estimand. Thirteen report tests of baseline balance and their "significance". In three the result of that test determined whether the analysis was adjusted. No paper reports a rule for outliers. The published baseline handling was analysis of covariance in six, a change score in four, the follow-up value in two and another approach in four. Model families were analysis of variance (five), mixed models (six), analysis of covariance (three), a t -test and a non-parametric test. Seven papers had no missing primary outcomes. Of the remaining nine, four analysed complete cases, three used likelihood-based longitudinal models, one multiple imputation and one did not say. Two papers modelled clustering. A protocol was available for seven trials, a registration only for six, neither for three. Ten trials appeared in journals that require data sharing. For six the requirement was unclear.

My first mapping of published analyses onto deposited data, made study by study before the rules of section 2.3 were fixed, differed from the rule-based mapping in 13 of the 16 trials: the control arm was coded differently in 10, the outcome variable in 5, the covariate set in 11, the baseline variable in 3 and the data file in 1 (Supplement Table S2). Most control-arm differences arose where the first value of the group variable had been taken as the control arm without checking value labels or group sizes against the paper. Running the first mapping through the final pipeline gave a median $|d|$ of 0.27 in both and a median estimand-preserving range of 0.09 against 0.12, but the sign of the median d was reversed in 7 of 16 trials, in each case because the control arm had been miscoded. The mapping errors changed the direction of effects, not their fragility.

4 Discussion

4.1 Principal findings

Of 152 trials in exercise, rehabilitation and physiotherapy whose availability statements pointed to a repository, 16 allowed a reader to identify and recompute the published primary analysis without contacting anyone. The loss occurred in two steps, first because deposits could not be reached or read, then because public data did not contain what the published analysis needed. Where the published analysis could be fitted as described, its estimate was recovered in 11 of 12 trials and to 0.014 standard deviations in all 12. Over choices that leave the estimand unchanged, the published result moved by at most 0.15 standard deviations in the typical trial and by 0.41 at most. It never changed sign. In 11 of 16 trials the only such choice that could move it was the handling of the baseline value. With that choice fixed the range was zero in 12 trials. Adding choices that change the estimand widened the range by about a third, mostly through logarithmic transformation. Three of the 12 primary reports do not identify a primary outcome through registration or paper, four papers give no point estimate, and one states an estimand. My own first mapping of published analyses onto deposited data reversed the direction of the effect in 7 of 16 trials before rules were fixed.

4.2 Two losses, one of them new

The first loss is the one earlier audits describe. Deposits behind truncated links, archives without a data file and studies that were not trials account for the drop from 152 to 48. Its size is in line with the availability rates of Jabouille and colleagues and the compliance rates of Elghzali and colleagues [7, 8]. The second loss is different in kind. In 11 of 48 trials the deposit could not support the published primary analysis: in eight the data were individual and readable but the allocation was not in the file, the primary outcome was not identifiable among the columns, the primary timepoint was missing, or the outcome existed only as raw signals. In three the deposit held a README, a summary table or an unreadable file. All eleven satisfy every availability policy I know of and are useless for the one purpose that justifies the policy. A deposit that can answer its own paper's question needs three things that none of these policies require: a variable that holds the allocation with labelled values, a variable that the paper names as the primary outcome, and a dictionary that connects the two to the text.

4.3 Reproduction

Eleven of 12 published estimates were recovered (Wilson interval 65 to 99%), ten of them to the printed decimal. None differed by more than 0.014 standard deviations. Every recomputed interval contains the published estimate. The interval on the proportion is wide because the denominator is 12. The denominator is 12 because only 12 of 423 screened records reach this step. The comparison with Brodeur and colleagues is one of setting, not of rate: they worked under a mandate with 110 articles, I without one. Naudet and colleagues reached the same conclusion as the original authors in 16 of 17 reanalyses [3]. The one discrepancy comes from a Bayesian model whose paper states no priors, not from the data. Reproduction required reading each paper closely and resolving four details the papers leave open. In one trial the authors' deposited code resolved them. Where a description was complete, a reader could recover

the number.

4.4 Fragility and what the grid could vary

Over choices that leave the estimand unchanged, the published result moved by at most 0.15 standard deviations in the typical trial and never changed sign. The number describes one choice. In 11 of 16 trials the only such choice that could move the estimate was the handling of the baseline value, because the covariates of the published model were not in the deposit, no cluster variable was estimable and the inference method changes the interval and not the estimate. The range over the three baseline analyses is not a property of analytic freedom. It is a function of the baseline imbalance that randomisation happened to produce, with a rank correlation of 0.82 over 15 trials, as the difference between change-score and covariance-adjusted estimates under imbalance predicts [38, 39]. With baseline handling fixed to analysis of covariance, which has justified precedence, the range was zero in 12 trials and at most 0.04 standard deviations in the other four. For simple two-arm trials with one primary outcome and one follow-up, a multiverse over strictly equivalent choices has almost nothing to vary.

What a wider multiverse adds is a change of question. The defensible grid exceeded the estimand-preserving grid by a median factor of 1.35. The node that most often drove the additional range was logarithmic transformation, which estimates a ratio of geometric means. Veltri's simulation attributed three quarters of the estimate variance to preprocessing [18]. In these trials, once options that change the estimand are set apart, preprocessing choices that keep it contributed little. The outlier arm shows the other way in which an unclassified multiverse manufactures fragility: a rule of three median absolute deviations, applied to a percentage with a ceiling, moved one estimate by 1.08 standard deviations. No trial in the corpus used such a rule. Crenshaw and colleagues showed what a multiverse looks like for one trial [19]. Applying one classified grid to 16 trials shows how little of it survives classification.

Two cautions apply. In 11 of 16 trials the preprocessing pipelines left the data almost unchanged. Sign stability in those trials is not evidence of anything. And 196 estimand-preserving specifications over 16 trials, between 3 and 35 per trial, is a description of one small, self-selected corpus, not of a field. Five of the trials randomised fewer than 30 participants.

4.5 Reporting

The reproduction rate is bounded above by what the papers and their registrations state. Among the 12 primary reports, the registered primary outcome could be identified for eight. Three trials were never registered. For one registered trial neither the registry nor the paper names a primary outcome. Three registrations were made after recruitment had started. Four of the 16 papers report a p-value or an F statistic without an estimate of the arm difference. One states what quantity it is estimating. CONSORT has asked for a defined primary outcome and an effect size with its precision since 2010 [55]. The 2025 revision asks for the metric and the timepoint of every outcome [56]. The estimands framework asks for the quantity itself [57, 58]. Thirteen papers test baseline balance, a practice Senn argued against in 1994 [59]. Three let the test decide the adjustment, which makes the analysis depend on a random event. None reports what it did with outliers. The four secondary reports are a reminder that the outcome a paper analyses and the outcome its trial registered are not the same thing. The registration is where the

difference shows.

4.6 Auditor degrees of freedom

Mapping a published analysis onto a deposited dataset is itself an analysis with choices: which column is the outcome, which value is the control arm, which timepoint, which covariates. My first mapping, made study by study, differed from the rule-based mapping in 13 of 16 trials and reversed the direction of the effect in seven. Rerunning it through the final pipeline changed neither the median size of the effects nor their range by much. It changed their sign. These were coding errors, not choices between defensible alternatives. They belong to the reproducer, not to the trials. Brodeur and colleagues let reproducers choose their robustness checks and counted the results. The many-analysts studies show what such freedom does to conclusions [22, 24]. Fixed rules, written before the mapping and applied to every trial, are the remedy that was applied. The error rate of the first pass belongs in the report.

4.7 Limitations

The corpus is self-selected. Trials that share data are larger, more recent and better conducted than trials that do not [60]. Wicherts and colleagues found more reporting errors and weaker evidence among authors who did not share their data [61]. The association with errors was not confirmed in a conceptual repetition [62] and failed a replication [63]. The reproduction rate is therefore plausibly an upper bound for the field. No method extrapolates from this corpus to trials that did not share.

Sixteen trials, five of which randomised fewer than 30 participants and one without a baseline measurement, support statements about a typical case and a maximum, not about a distribution. Only continuous primary outcomes were analysed. Crossover and within-person designs were excluded. Four of the 16 papers analyse an outcome that their trial registered as secondary. Four trials name co-primary outcomes of which I analysed the first. Three trials compare two arms drawn from a three-arm design, with the standardising denominator computed from those two arms. In one trial the paper is inconsistent about which arm is the control. I followed the design section. The corpus came from one database with one search string and required a repository link in the availability statement. Trials whose data are published as a journal supplementary file, which Jabouille and colleagues count as directly available, were not captured. A second route through Zenodo and Dryad found 127 study-like deposits that were not matched to articles.

The audit was not preregistered, the mapping rules were fixed after the published analyses had been read, and one coder mapped the trials. The estimand-preserving grid keeps a node whose options differ in precision. In most trials it was the only node that could vary. The strict grid shows what remains without it. The model family was inert: in the cluster-randomised FREE trial [29], random intercepts for practitioner and practice were singular in every fit. The linear-model specifications, including permutation inference over individual allocations, do not respect the cluster design. The variance partition is additive and treats confounded node pairs as separate. The named component of the integrity screen could be computed in one trial. Instrument-specific minimal important differences were not collected. The 0.2 standard deviation threshold is a convention. Pooled estimation over specifications and multiplicity control over specifications were not attempted [45, 46].

4.8 Implications

For journals and repositories, a deposit should be accepted when it contains a labelled allocation variable, a variable named as the primary outcome, and a dictionary. Without them the deposit meets the policy and fails its purpose. For multiverse practice, classify every decision before computing, drop options the literature does not use, collapse specifications that give the same result, fix the standardisation denominator before preprocessing, and report how many nodes could vary in each trial, because a multiverse over simple two-arm trials may have little to vary. For meta-research in fields whose journals do not require data deposition, 16 of 152 trials with a repository link (11%, Wilson interval 7 to 16%) reached recomputation and 11 of 152 (7%, 4 to 12%) reproduced. When the published analysis could be fitted as described, it reproduced.

5 Conclusion

In a field whose journals do not require data deposition, 16 of 152 trials with a repository link, about one in nine, allowed a reader to recompute the published primary analysis without contacting the authors. The loss occurred in two steps. The second, from public data to identifiable analysis, is the one that availability policies do not address. Where the published analysis could be fitted as described, 11 of 12 estimates were recovered and all 12 to 0.014 standard deviations. Over choices that leave the estimand unchanged, the published result moved by at most 0.15 standard deviations in the typical trial and never changed sign. In most trials the only such choice was the handling of the baseline value. Without it there was almost nothing to vary. Choices that change the estimand widened the range by a third. Three of 12 primary reports do not identify a primary outcome through registration or paper. Four papers give no estimate. My own first mapping of analyses onto data reversed seven effects before rules were fixed.

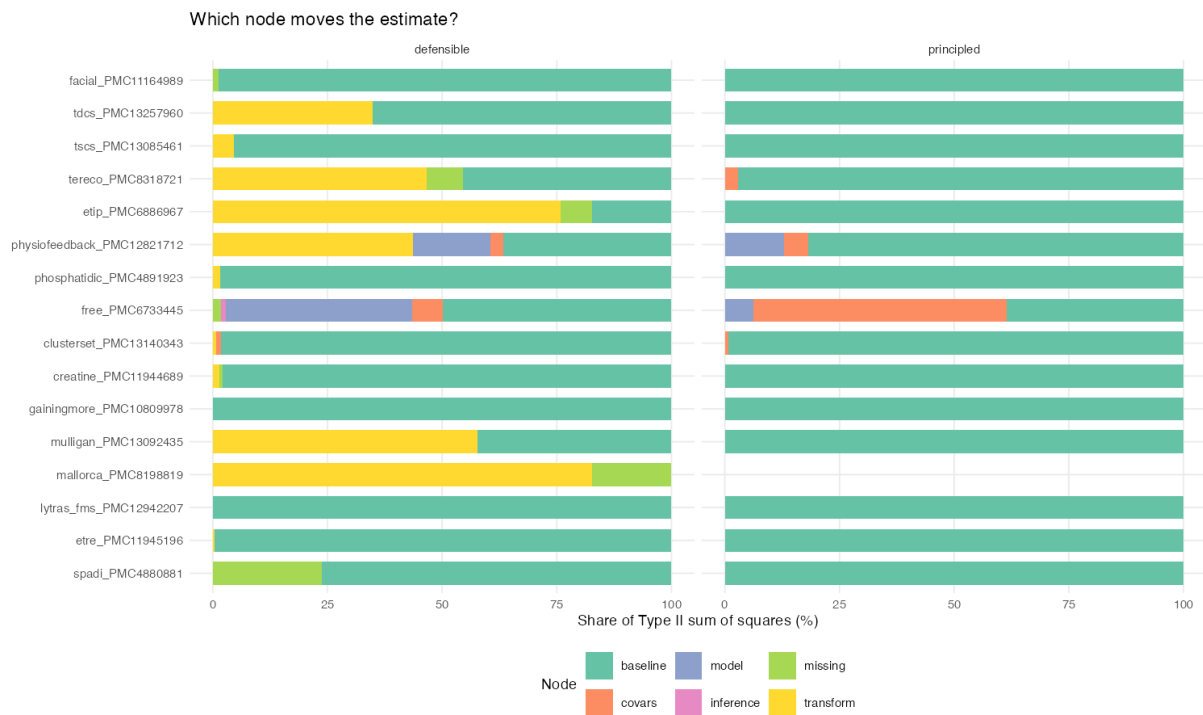


Figure 4: Share of the variance of d attributable to each decision node per trial (Type II sums of squares, additive model), defensible and estimand-preserving grid. Trials in which only one node varies show a share of 100% for that node by construction.

Tables

Table 1. Decision nodes of the grids

Node	Type	Options	Applicable when	Justification
Baseline handling	E on estimand, N on precision (see Methods 2.6)	endpoint, change, ancova	baseline measurement in deposit, including long-format deposits	Under randomisation all three are unbiased for the same mean difference; analysis of covariance is the most precise [37–39]
Covariate set	E	none, strat, published	stratification factors or published covariates in deposit	Adjustment is an efficiency choice, not a confounding correction, for a continuous outcome in a linear model [40, 41]
Outlier winsorising, pooled over arms	U (sensitivity arm only)	none, sd3_winsor, iqr15_winsor, mad_winsor	rule changes at least one value	No paper in the corpus reports an outlier rule; winsorising changes the quantity estimated; reported as a separate arm
Model family	E	lm, mixed_ri	estimable cluster variable	Same estimand; random intercept only where a cluster variable with three or more levels exists
Inference	E	model, hc3, permutation	always (HC3 and permutation only with the linear model)	Same estimand, different calibration of the interval
Missing data and analysis population	N	complete_case, mi, locf	missing primary outcomes present; LOCF only with a baseline	Multiple imputation has justified precedence under a missing-at-random assumption; last observation carried forward assumes no change after dropout [42]
Transformation	N	none, log	all outcome and baseline values positive	Log scale estimates a ratio of geometric means, not a difference
Outlier winsorising, arm-wise	U (sensitivity arm only)	pooled, armwise	an outlier rule is applicable and arm-wise limits differ from pooled	Arm-wise limits use post-randomisation information

Estimand-preserving grid: baseline handling, covariate set, model family, inference. Strict grid: the same with baseline handling fixed to analysis of covariance. Defensible grid: estimand-preserving grid plus missing-data strategy and transformation. Outlier arm: estimand-preserving grid plus the two outlier nodes, sensitivity only. Incoherent combinations removed; exact duplicates collapsed.

Table 2. Reporting of the primary analysis in the 16 included trials

Trial	Year	Report type	n randomised	Registration	Primary outcome declared in	Point estimate	Estimand stated	Baseline test	Baseline handling	Covariates	Missing data	Model	Journal requires sharing
Balance feedback [30]	2026	secondary	404	yes	secondary report	yes	no	yes	other	published_set	mmrm_ml	mixed_ri	unclear
Mulligan, neck pain [49]	2026	primary	43	yes	registration	no	no	yes	change	none	none_missing	anova	unclear
Lytras, low back pain [50]	2026	primary	40	yes, after start	registration	yes	no	yes	other	none	none_missing	anova	yes
Cluster sets, squat 1RM [53]	2026	primary	36	none found	sample-size calculation only registration	no	no	yes, decisive	other	published_set	none_missing	mixed_ri	unclear
TERECO, COVID-19, 6MWD [31]	2022	primary	120	yes	registration	yes	no	no	ancova	baseline_plus_estimand	mmrm_ml	mixed_ri	yes
Frozen shoulder, SPADI [34]	2016	primary	106	yes, after start	registration	yes	no	yes	ancova	baseline_only	not_reported	anova	unclear

Trial	Year	Report type	n randomised	Registration	Primary outcome declared in	Point estimate	Estimand stated	Baseline test	Baseline handling	Covariates	Missing data	Model	Journal requires sharing
tSCS, balance, BBS [51]	2026	primary	20	yes	registration	yes	no	yes, decisive	ancova	baseline_only	none_missing	lm_ancova	yes
tDCS after ACLR, CAR [48]	2026	secondary	20	yes	secondary report	yes	no	yes	endpoint	none	none_missing	ttest	yes
ETIP, pregnancy, PGWBI [47]	2019	secondary	91	yes	secondary report	yes	no	yes	ancova	baseline_only	complete_cas	dm_ancova	yes
Phosphatidic acid, LBM [28]	2016	primary	18	none found	not declared	no	no	yes	change	none	none_missing	anova	yes
Facial palsy, SFGS [52]	2024	primary	24	yes	registration	no	no	yes	change	none	complete_cas	nonparam	yes
Deload, muscle thickness [32]	2024	primary	50	yes	not declared	yes	yes	no	change	none	complete_cas	mixed_ri	yes
eTRE, body weight [27]	2025	primary	24	none found	paper only	yes	no	yes	other	none	none_missing	mixed_ri	yes
Creatine, lean mass [54]	2025	primary	63	yes	registration	yes	no	yes	ancova	baseline_only	complete_cas	dm_ancova	unclear

Trial	Year	Report type	n randomised	Registration	Primary outcome declared in	Point estimate	Estimand stated	Baseline test	Baseline handling	Covariates	Missing data	Model	Journal requires sharing
FREE, low back pain, RMDQ [29]	2019	primary	226	yes	registration	yes	no	no	ancova	published_set	mrmr_ml	mixed_slopes	yes
Water exercise, EPDS [33]	2021	secondary	294	yes, after start	secondary report	yes	no	yes, decisive	endpoint	none	mi	anova	unclear

Table 3. Computational reproduction of the published primary estimate, published analysis fitted as described

Trial	n in fitted model	Published estimate (95% CI; p)	Recomputed estimate (95% CI; p)	Relative difference	Absolute difference (SD)	Class	Recomputed CI contains published	Published estimate: percentile in estimand-preserving multiverse	Grid matches tied
Creatine, lean mass [54]	63	0.51 (CI not reported; p = 0.03)	0.507 (0.0569 to 0.958; p = 0.028)	0.5%	0.000	full	yes	0.33	1
ETIP, pregnancy, PGWBI [47]	54	2.6 (-3.77 to 8.97; p = 0.42)	2.6 (-3.77 to 8.97; p = 0.42)	0.1%	0.000	full	yes	0.67	1
eTRE, body weight [27]	24	-2.13 (-3.07 to -1.18; p = 0.001)	-2.13 (-3.07 to -1.18; p = 0.00024)	0.2%	0.002	full	yes	0.33	1

Trial	n in fitted model	Published estimate:							
		Published estimate (95% CI; p)	Recomputed estimate (95% CI; p)	Relative difference	Absolute difference (SD)	Class	Recomputed CI contains published	percentile in estimate-preserving multiverse	Grid matches tied
FREE, low back pain, RMDQ [29]	221	0.57 (-0.64 to 1.78; p = 0.35)	0.58 (-0.7 to 1.86; p = 0.37)	1.8%	0.002	full	yes	1.00	2 (range 0.01 SD)
Lytras, low back pain [50]	40	-1.2 (-2.18 to -0.22; p = 0.018)	-1.2 (-2.18 to -0.222; p = 0.018)	0.0%	0.000	full	yes	0.67	3 (range 0.23 SD)
Water exercise, EPDS [33]	294	-0.7 (-1.24 to -0.17; p = 0.01)	-0.694 (-1.23 to -0.161; p = 0.011)	0.8%	0.003	full	yes	0.00	1
Balance feedback [30]	373	0.02 (-0.08 to 0.12; p = 0.68)	0.021 (-0.0782 to 0.12; p = 0.68)	4.9%	0.002	full	yes	not on common scale	2 (range 0.05 SD)
Frozen shoulder, SPADI [34]	70	-20.8 (-28.9 to -12.7; p not reported)	-20.8 (-28.9 to -12.7; p = 0.0000026)	0.1%	0.001	full	yes	0.67	1
tDCS after ACLR, CAR [48]	20	6.85 (1.92 to 11.8; p = 0.01)	6.85 (2.27 to 11.4; p = 0.0056)	0.0%	0.000	full	yes	0.33	1
TERECO, COVID-19, 6MWD [31]	119	65.5 (43.8 to 87.1; p not reported)	65.4 (43.8 to 87.1; p = 0.0000000091)	0.0%	0.000	full	yes	0.50	1
tSCS, balance, BBS [51]	20	7.67 (5.14 to 10.2; p not reported)	7.67 (5.14 to 10.2; p = 0.0000064)	0.1%	0.000	full	yes	0.33	1
Deload, muscle thickness [32]	39	-0.63 (-2.8 to 1.5; p = 0.27)	-0.515 (-2.73 to 1.76; posterior model)	18.3%	0.014	major	yes	0.67	1

Trial	n in fitted model	Published estimate:							
		Published estimate (95% CI; p)	Recomputed estimate (95% CI; p)	Relative difference	Absolute difference (SD)	Class	Recomputed CI contains published	percentile in estimand-preserving multiverse	Grid matches tied
Cluster sets, squat 1RM [53]	36	none (interaction p = 0.352; F time 80.67, F sex 95.13)	interaction p = 0.352; F 80.67, 95.13			no point estimate; statistics reproduce			
Facial palsy, SFGS [52]	23	none (Mann-Whitney p = 0.002; medians 59.0 vs 24.0)	exact p = 0.0020; medians 59.0 vs 24.0			no point estimate; statistics reproduce			
Mulligan, neck pain [49]	43	none (group x time F = 0.983, p = 0.383)	F(2,40) = 0.983, p = 0.383			no point estimate; statistics reproduce			
Phosphatidic acid, LBM [28]	18	none (interaction F(1,16) = 33.30, p = 0.041)	F(1,16) = 4.95, p = 0.041			no point estimate; p reproduces, F misreported			

Full: same sign and relative difference at most 2%, or within half a unit of the last printed digit; minor: at most 10%; major: otherwise (rule fixed before computation). Assumptions made where the paper is silent are recorded in `R/stage2_published_models.R`; classification in `R/stage2_classify.R`.

References

- [1] Brodeur A, Mikola D, Cook N, Fiala L, Brailey T, Briggs R, et al. Reproducibility and robustness of economics and political science research. *Nature*. 2026;652(8108):151–156.
- [2] Stodden V, Seiler J, Ma Z. An empirical analysis of journal policy effectiveness for computational reproducibility. *Proceedings of the National Academy of Sciences*. 2018;115(11):2584–2589.
- [3] Naudet F, Sakarovitch C, Janiaud P, Cristea I, Fanelli D, Moher D, et al. Data sharing and reanalysis of randomized controlled trials in leading biomedical journals with a full data sharing policy: survey of studies published in *The BMJ* and *PLOS Medicine*. *BMJ*. 2018:k400.
- [4] Siebert M, Gaba J, Renault A, Laviolle B, Locher C, Moher D, et al. Data-sharing and re-analysis for main studies assessed by the European Medicines Agency—a cross-sectional study on European Public Assessment Reports. *BMC Medicine*. 2022;20(1).
- [5] Hardwicke TE, Mathur MB, MacDonald K, Nilsson G, Banks GC, Kidwell MC, et al. Data availability, reusability, and analytic reproducibility: evaluating the impact of a mandatory open data policy at the journal *Cognition*. *Royal Society Open Science*. 2018;5(8):180448.
- [6] Hardwicke TE, Bohn M, MacDonald K, Hembacher E, Nuijten MB, Peloquin BN, et al. Analytic reproducibility in articles receiving open data badges at the journal *Psychological Science* : an observational study. *Royal Society Open Science*. 2021;8(1).
- [7] Elghzali A, Molina D, Chaudhry M, Duncan J, Elfar A, Oldham E, et al. Data sharing practices in high-impact rehabilitation journals. *BMC Medical Research Methodology*. 2025;25(1).
- [8] Jabouille F, Farajzadeh A, Exertier AG, Martine J, Sadek N, Klinger LH, et al. Replicability and transparency in physical therapy research: Time to wake up. *medRxiv*. 2025.
- [9] Murphy J, Caldwell AR, Mesquida C, Ladell AJM, Encarnación-Martínez A, Tual A, et al. Estimating the Replicability of Sports and Exercise Science Research. *Sports Medicine*. 2025;55(10):2659–2679.
- [10] Dhanani NH, Olavarria OA, Bell CS, Holihan JL, Liang MK. Reproducing Statistics Performed in Published Randomized Controlled Trials. *Journal of General Internal Medicine*. 2022;37(16):4197–4201.
- [11] Nolte S. Computational Reproducibility in Sports Science: A Registered Report Reanalyzing Private Raw Data. *OSF Registries*. 2026.
- [12] Nolte S, Memmert D, Rein R. Data and Code Availability in Sports Science: A Registered Report. *OSF Preprints*. 2026.
- [13] Steegen S, Tuerlinckx F, Gelman A, Vanpaemel W. Increasing Transparency Through a Multiverse Analysis. *Perspectives on Psychological Science*. 2016;11(5):702–712.
- [14] Simonsohn U, Simmons JP, Nelson LD. Specification curve analysis. *Nature Human Behaviour*. 2020;4(11):1208–1214.
- [15] Nepomuceno A, Ghosal A, Sandoval-Lentisco A, Ioannidis JPA. Uptake and Implementation of Multiverse-style Analyses Across 613 Studies. *openRxiv preprint*. 2026.

- [16] Del Giudice M, Gangestad SW. A Traveler's Guide to the Multiverse: Promises, Pitfalls, and a Framework for the Evaluation of Analytic Decisions. *Advances in Methods and Practices in Psychological Science*. 2021;4(1).
- [17] Short CA, Breznau N, Bruntsch M, Burkhardt M, Busch NA, Cesnaite E, et al. Multicurious: A Multidisciplinary Guide to Multiverse Analysis. *Advances in Methods and Practices in Psychological Science*. 2026;9(2).
- [18] Veltri GA. The Effects of Data Preprocessing Choices on Behavioral RCT Outcomes: A Multiverse Analysis. *Multivariate Behavioral Research*. 2025;61(2):383–398.
- [19] Crenshaw AO, Pukay-Martin ND, Wagner AC, Schobitz RP, Yarvis JS, Young-McCaughan S, et al. Navigating Analytical Challenges in Clinical Trials Using the Multiverse Approach. *Collabra: Psychology*. 2026;12(1).
- [20] Patel CJ, Burford B, Ioannidis JPA. Assessment of vibration of effects due to model specification can demonstrate the instability of observational associations. *Journal of Clinical Epidemiology*. 2015;68(9):1046–1058.
- [21] Klau S, Hoffmann S, Patel CJ, Ioannidis JP, Boulesteix AL. Examining the robustness of observational associations to model, measurement and sampling uncertainty with the vibration of effects framework. *International Journal of Epidemiology*. 2020;50(1):266–278.
- [22] Silberzahn R, Uhlmann EL, Martin DP, Anselmi P, Aust F, Awtrey E, et al. Many Analysts, One Data Set: Making Transparent How Variations in Analytic Choices Affect Results. *Advances in Methods and Practices in Psychological Science*. 2018;1(3):337–356.
- [23] Botvinik-Nezer R, Holzmeister F, Camerer CF, Dreber A, Huber J, Johannesson M, et al. Variability in the analysis of a single neuroimaging dataset by many teams. *Nature*. 2020;582(7810):84–88.
- [24] Breznau N, Rinke EM, Wuttke A, Nguyen HHV, Adem M, Adriaans J, et al. Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty. *Proceedings of the National Academy of Sciences*. 2022;119(44).
- [25] Gould E, Fraser HS, Parker TH, Nakagawa S, Griffith SC, Vesik PA, et al. Same data, different analysts: variation in effect sizes due to analytical decisions in ecology and evolutionary biology. *BMC Biology*. 2025;23(1).
- [26] Van den Akker OR, Weston S, Campbell L, Chopik B, Damian R, Davis-Kean P, et al. Preregistration of secondary data analysis: A template and tutorial. *Meta-Psychology*. 2021;5.
- [27] Yu Z, Ueda T. Early Time-Restricted Eating Improves Weight Loss While Preserving Muscle: An 8-Week Trial in Young Women. *Nutrients*. 2025;17(6):1022.
- [28] Escalante G, Alencar M, Haddock B, Harvey P. The effects of phosphatidic acid supplementation on strength, body composition, muscular endurance, power, agility, and vertical jump in resistance trained men. *J Int Soc Sports Nutr*. 2016;13:24.

- [29] Darlow B, Stanley J, Dean S, Abbott JH, Garrett S, Wilson R, et al. The Fear Reduction Exercised Early (FREE) approach to management of low back pain in general practice: A pragmatic cluster-randomised controlled trial. *PLoS Med.* 2019;16(9):e1002897.
- [30] Suarez JRM, Lafontant K, Banarjee C, Xie R, Park JH, Thiamwong L. Effect of a Physio-Feedback Exercise Intervention Program on the Static Balance of Community-Dwelling Older Adults: A Clustered Randomized Controlled Trial. *Geriatrics (Basel).* 2026;11(1):6.
- [31] Li J, Xia W, Zhan C, Liu S, Yin Z, Wang J, et al. A telerehabilitation programme in post-discharge COVID-19 patients (TERECO): a randomised controlled trial. *Thorax.* 2022;77(7):697-706.
- [32] Coleman M, Burke R, Augustin F, Piñero A, Maldonado J, Fisher JP, et al. Gaining more from doing less? The effects of a one-week deload period during supervised resistance training on muscular adaptations. *PeerJ.* 2024;12:e16777.
- [33] Navas A, Carrascosa MDC, Artigues C, Ortas S, Portells E, Soler A, et al. Effectiveness of Moderate-Intensity Aerobic Water Exercise during Pregnancy on Quality of Life and Postpartum Depression: A Multi-Center, Randomized Controlled Trial. *J Clin Med.* 2021;10(11):2432.
- [34] Sharma SP, Bærheim A, Moe-Nilssen R, Kvåle A. Adhesive capsulitis of the shoulder, treatment with corticosteroid, corticosteroid with distension or treatment-as-usual; a randomised controlled trial in primary care. *BMC Musculoskelet Disord.* 2016;17:232.
- [35] Carlisle JB. Data fabrication and other reasons for non-random sampling in 5087 randomised, controlled trials in anaesthetic and general medical journals. *Anaesthesia.* 2017;72(8):944–952.
- [36] Bolland MJ, Gamble GD, Avenell A, Grey A. Rounding, but not randomization method, non-normality, or correlation, affected baseline P-value distributions in randomized trials. *Journal of Clinical Epidemiology.* 2019;110:50–62.
- [37] Vickers AJ, Altman DG. Analysing controlled trials with baseline and follow up measurements. *BMJ.* 2001;323(7321):1123–1124.
- [38] Senn S. Change from baseline and analysis of covariance revisited. *Statistics in Medicine.* 2006;25(24):4334–4344.
- [39] Van Breukelen GJP. ANCOVA versus change from baseline had more power in randomized studies and more bias in nonrandomized studies. *Journal of Clinical Epidemiology.* 2006;59(9):920–925.
- [40] Kahan BC, Jairath V, Doré CJ, Morris TP. The risks and rewards of covariate adjustment in randomized trials: an assessment of 12 outcomes from 8 studies. *Trials.* 2014;15(1).
- [41] Kahan BC, Morris TP. Reporting and analysis of trials using stratified randomisation in leading medical journals: review and reanalysis. *BMJ.* 2012;345(sep14 1):e5840–e5840.
- [42] White IR, Royston P, Wood AM. Multiple imputation using chained equations: Issues and guidance for practice. *Statistics in Medicine.* 2010;30(4):377–399.
- [43] Rafi Z, Greenland S. Semantic and cognitive tools to aid statistical science: replace confidence and significance by compatibility and surprise. *BMC Medical Research Methodology.* 2020;20(1).

- [44] Wasserstein RL, Schirm AL, Lazar NA. Moving to a World Beyond “ $p < 0.05$ ”. *The American Statistician*. 2019;73(sup1):1–19.
- [45] Bartoš F, Hoogeveen S, Sarafoglou A, Pawel S. Single-Dataset Meta-Analysis For Many-Analysts And Multiverse Studies. *arXiv preprint arXiv:251117064*. 2025.
- [46] Mandl MM, Becker-Pennrich AS, Hinske LC, Hoffmann S, Boulesteix AL. Addressing researcher degrees of freedom through minP adjustment. *BMC Medical Research Methodology*. 2024;24(1).
- [47] Garnæs KK, Helvik AS, Stafne SN, Mørkved S, Salvesen K, Salvesen et al. Effects of supervised exercise training during pregnancy on psychological well-being among overweight and obese women: secondary analyses of the ETIP-trial, a randomised controlled trial. *BMJ Open*. 2019;9(11):e028252.
- [48] Esfahani NH, Roostayi MM, Hovsepian V, Baghban AA, Rezaeian ZS, Jaberzadeh S. Integrating multi-session transcranial direct current stimulation with routine physical therapy to improve quadriceps strength and activation in athletes during subacute recovery following ACL reconstruction: A double-blind RCT. *PLoS One*. 2026;21(6):e0345947.
- [49] Kiouloukiotis I, Stathis E, Iakovidis P, Kasimis K, Georgoulas V, Chatziprodromidou IP, et al. Comparison of the short-term effects of Mulligan and Maitland mobilisation techniques combined with home-based exercise for non-specific neck pain: A randomised controlled trial. *Hong Kong Physiother J*. 2026;46(1):67-82.
- [50] Lytras D, Iakovidis P, Kasimis K, Georgoulas V, Algiounidis I, Kamparoudi GM, et al. Short-Term Effects of Manual Therapy Combined with Functional Magnetic Stimulation in Individuals with Lumbar Disk Herniation with Radiculopathy: A Randomized Clinical Trial. *Medicina (Kaunas)*. 2026;62(2):249.
- [51] Rashad A, Xijing H, Hui W, Xuefei H, Zhenyu X, Yan J, et al. Transcutaneous spinal cord stimulation combined with robot-assisted gait training to improve mobility and balance in spinal cord injury: intramuscular electromyography as a biomarker for functional recovery. *BMC Med*. 2026;24(1):245.
- [52] Santiago S, Joshua AM, Nayak A, Misri Z, Pai S, Pai R, et al. Effectiveness of novel facial stretching with structured exercise versus conventional exercise for Bell’s palsy: a single-blinded randomized clinical trial. *Sci Rep*. 2024;14(1):13266.
- [53] Hobein E, Miltner F, Jukic I, Ferrauti A, Wiewelhove T. Acute Responses and Chronic Adaptations to Cluster Versus Traditional Set Resistance Training in Males and Females. *Eur J Sport Sci*. 2026;26(4):e70160.
- [54] Desai I, Pandit A, Smith-Ryan AE, Simar D, Candow DG, Kaakoush NO, et al. The Effect of Creatine Supplementation on Lean Body Mass with and Without Resistance Training. *Nutrients*. 2025;17(6):1081.
- [55] Schulz KF, Altman DG, Moher Da. CONSORT 2010 Statement: updated guidelines for reporting parallel group randomised trials. *BMJ*. 2010;340(mar23 1):c332–c332.
- [56] Hopewell S, Chan AW, Collins GS, Hróbjartsson A, Moher D, Schulz KF, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. *BMJ*. 2025;389:e081123.

- [57] International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. ICH E9(R1): Addendum on Estimands and Sensitivity Analysis in Clinical Trials to the Guideline on Statistical Principles for Clinical Trials. ICH; 2019. Step 4 version, 20 November 2019. Available from: https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf.
- [58] Kahan BC, Hindley J, Edwards M, Cro S, Morris TP. The estimands framework: a primer on the ICH E9(R1) addendum. *BMJ*. 2024:e076316.
- [59] Senn S. Testing for baseline balance in clinical trials. *Statistics in Medicine*. 1994;13(17):1715–1726.
- [60] Tsujimoto Y, Fujii T, Onishi A, Omae K, Luo Y, Imai H, et al. No consistent evidence of data availability bias existed in recent individual participant data meta-analyses: a meta-epidemiological study. *Journal of Clinical Epidemiology*. 2020;118:107–114.e5.
- [61] Wicherts JM, Bakker M, Molenaar D. Willingness to Share Research Data Is Related to the Strength of the Evidence and the Quality of Reporting of Statistical Results. *PLoS ONE*. 2011;6(11):e26828.
- [62] Nuijten MB, Borghuis J, Veldkamp CLS, Dominguez-Alvarez L, van Assen MALM, Wicherts JM. Journal Data Sharing Policies and Statistical Reporting Inconsistencies in Psychology. *Collabra: Psychology*. 2017;3(1).
- [63] Claesen A, Vanpaemel W, Maerten AS, Verliefde T, Tuerlinckx F, Heyman T. Data sharing upon request and statistical consistency errors in psychology: A replication of Wicherts, Bakker and Molenaar (2011). *PLOS ONE*. 2023;18(4):e0284243.